

Performance of Combined Surface and Intramuscular EMG for Classification of Hand Movements

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Abstract— The surface EMG (sEMG) has been used as control source for upper limb prosthetics since decades. Previous studies suggested that intramuscular EMG showed promising results for upper limb prosthetics. This study investigates the strength of combined surface and intramuscular EMG (cEMG) for improved myoelectric control. Five able-bodied subjects and three transradial amputees were evaluated using offline classification error as performance metric. Six surface and intramuscular channels were recorded concurrently from each subject for seven consecutive days and Stacked sparse autoencoders (SSAE) and LDA classifiers were used for classification. As a control source, either sEMG channels were used or combined channels were used with reduced features using PCA. In the within session analysis, cEMG ($2.21 \pm 1.19\%$) outperformed the sEMG ($4.63 \pm 2.07\%$) for both able-bodied and amputee subjects using SSAE. For between session analysis, cEMG outperformed the sEMG for both able-bodied and amputee subjects with percentage points difference of 7.93. These results imply cEMG can significantly improve the performance of pattern recognition based myoelectric control scheme for amputee subjects too and further improvement can be made by utilizing SSAE which show improved performance as compared to LDA.

I. INTRODUCTION

A myoelectric prosthetic hand is controlled by electrical activity that is produced within the remnant muscles and the technique for recording this electrical activity is EMG. EMG signals are recorded from a contracting muscle using either a wire, needle or surface electrodes. Surface EMG (sEMG) is the non-invasive [1] source for myoelectric control of upper limb prosthetics and is being more commonly used in the literature on myoelectric control [2]. While intramuscular EMG (iEMG) is an invasive technique [3] and comparing to sEMG, it has little crosstalk and provide stable control signals from deep muscles [3-5]. However, iEMG contains information from specific muscles and hence it may be a disadvantage unlike sEMG which include more local information from surrounding muscles [6].

Myoelectric control schemes have been investigated with both the sEMG and iEMG as control sources [7, 8]. Traditional schemes adopt simple strategies to encode the EMG signals

recorded from one or more sites [9]. E.g. on-off, direct and proportional schemes use the amplitude of EMG signals with some post-processing, but they are not capable enough to precisely control the multi-functionality of dexterous prosthetic hands [10, 11]. This limitation could be better addressed with pattern recognition (PR) based control strategies [12-15] which have shown promising results with both the sEMG and iEMG-based schemes. The main requirement of PR-based control is that patterns should be repeatable and differentiable from each other [16]. Two important steps of PR-based control are features extraction and classification. The extracted features from EMG signals provide comparatively more and specific information in order to utilize greater functionality of a prosthetic hand. Different classifiers have been evaluated for myoelectric control schemes including artificial neural networks, support vector machines, random forest and decision tree etc., but LDA comes out to be the more common and robust. Previous studies [2, 9, 17-20] based on myoelectric control schemes, suggest that either there was no significant difference found between iEMG and sEMG or the sEMG performed better than iEMG. Kamavuako et al. [21] performed a real-time study in order to evaluate the strength of combined sEMG and iEMG for improved myoelectric control. In that study, five performance metrics were used for real time and one as offline classification error (concern of this study). In real-time performance metrics, combined EMG (cEMG) either outperformed sEMG or there was no significant difference found, while in the offline study, cEMG performed significantly better than sEMG. However, that study was related to only able-bodied subjects and there is no guarantee about the repeatability of results with amputee subjects.

Therefore, the main goal of this study was to compare the performance of both combined and surface EMG signals using classification error as performance metric. It is evaluated using both SSAE and LDA classifiers.

II. METHODOLOGY

A. Data collection and Experimental procedure

Five able-bodied and three transradial amputee subjects participated in this study. For experimentation, right hand of

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all subjects was used except for two amputee subjects where their left hand was used. The experimental procedures were in accordance with the Declaration of Helsinki and were approved by the ethical committee of Riphah International University. Six surfaces and six intramuscular channels were recorded concurrently from each able-bodied subject. In case of amputees, two subjects had only five surface electrodes while the number of intramuscular electrodes varied from three to six due to limited space. The signals were sampled with 8 KHz. Both sEMG and iEMG were filtered with analogue bandpass filters of bandwidth 10-500 Hz and 0.1-4 KHz, respectively.

Each subject performed eleven movements for seven consecutive days and on each individual day, each movement was repeated four times with a contraction and relaxation time of five seconds. Movements included hand open, hand close, flex hand, extend hand, pronation, supination, side grip, fine grip, agree, pointer and rest. Each session took 400 secs to complete, and the sequence of movements was randomized for each session.

B. Data processing

Both the intramuscular and surface data was filtered with digital 3rd order Butterworth filter of bandwidths 0.06-1.5 KHz and 20-500 Hz, respectively. The electric noise was removed with a 3rd order configurable Butterworth filter and features were extracted from a 200 msec time window with an increment of 28.5 msec. Features included mean absolute value (MAV), waveform length (WL), zero crossing (ZC) and slope-sign change (SSC) [14]. Classification was performed with both SSAE [22] and LDA [23].

Two layers SSAE are used in this work having 24 and 12 units in both layer 1 and 2, respectively. Different tuning parameters including L2Regularization (L2R), sparsity proportion (SP) and sparsity regularization (SR) were

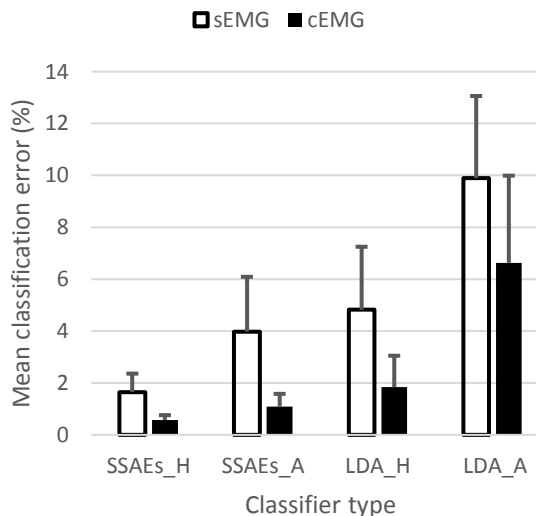


Figure 1. Within session analysis. Mean classification errors for both kinds of data obtained with SSAE and LDA. _H and _A represents the data for both healthy and amputee subjects respectively. bars represent the standard deviation

TABLE 1. P VALUES OBTAINED WITH FRIEDMAN'S TEST FOR BOTH CEMG AND SEMG DATA OF HEALTHY AND AMPUTEE SUBJECTS. IN THIS TEST, ONE FACTOR WAS THE EMG DATA TYPE AND ANOTHER FACTOR WAS THE NUMBER OF DAYS

Classifier	Mean accuracy $\pm$ standard deviation		P value
	sEMG	Combined EMG	
SSAE_H	98.35 $\pm$ 0.71	99.43 $\pm$ 0.19	0.0021
LDA_H	95.18 $\pm$ 2.43	98.16 $\pm$ 1.21	0.0019
SSAE_A	96.02 $\pm$ 2.11	98.91 $\pm$ 0.49	0.0001
LDA_A	90.11 $\pm$ 3.17	93.37 $\pm$ 3.36	0.0001

optimized for both layers. The final values (L2R= 0.0001, SR=0.01 and SP=0.5) were chosen for both layers.

In sEMG based control, all six surface channels were used while for combined EMG based control, features vector from all channels were reduced by using principal component analysis (PCA) and were equalized to that of surface features. For within session analysis, four-fold cross validation scheme was used while eight-fold cross validation was used in the between session analysis. All these steps were performed in exactly the same way for both healthy and amputee subjects. The same folds were used for classification using LDA and SSAEs to make a fair comparison.

In order to quantify the difference of sEMG vs combined EMG and SSAE vs LDA, Friedman's statistical test was used and P values less than 0.05 were considered significant

III. RESULTS

A. Within session analysis

In this analysis, for an individual subject, data of a single day were randomly divided into four-folds (each fold

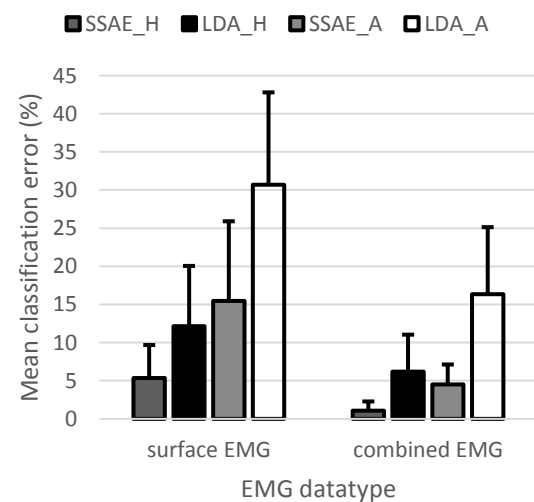


Figure 2. Between session analysis. Mean classification errors for both kinds of data obtained with SSAEs and LDA. _H and _A represents the data for both healthy and amputee subjects, respectively. Bars represent the standard deviation.

TABLE 2. P VALUES OBTAINED WITH FRIEDMAN'S TEST FOR BOTH CEMG AND SEMG DATA OF HEALTHY AND AMPUTEE SUBJECTS. IN THIS TEST, ONE FACTOR WAS THE EMG DATA TYPE AND ANOTHER FACTOR WAS THE NUMBER OF DAYS

Classifier	Mean accuracy $\pm$ standard deviation		P value
	sEMG	Combined EMG	
SSAE_H	94.63 $\pm$ 4.31	98.91 $\pm$ 1.19	0.0001
LDA_H	87.86 $\pm$ 7.91	93.80 $\pm$ 4.84	0.0000
SSAE_A	84.54 $\pm$ 10.44	95.48 $\pm$ 2.61	0.0000
LDA_A	69.31 $\pm$ 12.11	83.67 $\pm$ 8.80	0.0000

contained equal number of all movement types). Three folds were used for training and one for testing. This was repeated four times and each time, a new fold was used for testing. Classification errors of seven sessions were averaged for each subject and results are presented as mean of all healthy and amputee subjects separately.

Figure 1 shows the mean classification errors obtained with both SSAEs and LDA for both healthy and amputee subjects.

To compare the performance of both kinds of data, statistical tests were conducted and P-values are tabulated in table 1.

From table 1, it can be verified that cEMG outperformed the sEMG data for both healthy and amputee subjects. Further, comparing performance of SSAEs and LDA, it was found that SSAEs achieved comparatively less classification error rates for both healthy and amputee subjects and outperformed LDA

TABLE 3. CEMG VS SEMG DATA OF HEALTHY SUBJECTS. IN EACH MATRIC, THE UPPER TRIANGLE SHOWS THE MEAN ERRORS FOR CEMG WHILE LOWER TRIANGLE IS FOR SEMG.

Healthy subjects cEMG vs sEMG with SSAEs							
Days	D1	D2	D3	D4	D5	D6	D7
D1		0.96	1.09	0.94	0.92	0.82	1.07
D2	4.44		1.30	1.20	1.01	0.92	1.28
D3	5.21	5.95		1.32	1.27	1.12	1.63
D4	4.38	5.30	5.89		0.99	0.96	1.24
D5	5.24	5.37	6.23	5.17		0.85	1.03
D6	4.89	5.10	5.72	4.76	4.68		0.76
D7	6.03	5.80	6.46	5.60	5.61	4.84	
Healthy subjects cEMG vs sEMG with LDA							
Days	D1	D2	D3	D4	D5	D6	D7
D1		5.78	6.33	6.38	6.85	6.60	7.26
D2	10.57		5.51	5.88	6.30	5.89	6.58
D3	10.71	10.20		6.21	6.96	6.25	6.65
D4	11.65	11.61	11.35		5.97	6.11	6.71
D5	13.39	13.11	12.99	11.76		5.45	5.46
D6	13.02	13.23	12.88	12.24	11.54		4.78
D7	13.78	13.89	13.25	11.97	11.60	10.15	

TABLE 4. CEMG VS SEMG DATA OF AMPUTEE SUBJECTS. IN EACH MATRIC, THE UPPER TRIANGLE SHOWS THE MEAN ERRORS FOR CEMG WHILE LOWER TRIANGLE IS FOR SEMG.

Amputee subjects cEMG vs sEMG with SSAEs							
Days	D1	D2	D3	D4	D5	D6	D7
D1		4.93	5.13	5.53	4.76	4.91	4.72
D2	15.89		4.61	4.31	4.08	3.98	4.11
D3	17.77	15.73		5.11	4.06	5.35	4.16
D4	15.95	15.02	17.86		4.15	4.62	4.57
D5	16.81	14.26	17.35	14.99		3.93	3.89
D6	14.60	13.18	14.09	13.35	11.85		3.60
D7	13.82	12.99	14.97	13.49	12.15	10.04	
Amputee subjects cEMG vs sEMG with LDA							
Days	D1	D2	D3	D4	D5	D6	D7
D1		18.69	19.25	17.54	20.25	17.07	17.27
D2	34.24		16.76	15.99	15.86	14.88	13.98
D3	37.74	33.69		16.16	16.92	14.51	14.42
D4	31.84	31.32	33.92		15.21	14.21	13.57
D5	35.62	32.84	34.24	30.03		12.94	12.13
D6	31.05	29.95	30.45	27.08	23.84		10.33
D7	33.87	30.55	31.54	27.56	25.15	20.66	

with 6.81 and 9.08 percentage points for both cEMG and sEMG, respectively.

B. Between session analysis

For seven sessions, we have a total of twenty-one unique pairs of sessions. Eight-fold cross validation was used for each pair and classification errors were averaged over twenty-one pairs for each subject. Results are presented as mean classification errors of healthy and amputee subjects separately (as shown in figure 2).

Like for within session analysis, SSAE also performed comparatively better for between sessions data and outperformed LDA with percentage points of 19.41 for both healthy and amputee subjects. Using SSAEs, cEMG and sEMG data of both able-bodied and amputee subjects achieved close error rates for both the within and between sessions data with percentage points of 1.72 and 7.60, respectively, while this difference using LDA was 7.03 and 14.06, respectively. Unlike LDA, SSAE significantly improved the performance.

IV. DISCUSSION

The purpose of this study was to investigate the performance of SSAE in comparison with a conventional machine learning method and to evaluate whether the combined sEMG and iEMG from deep muscles could significantly improve the performance of myoelectric control for both able-bodied and transradial amputee subjects.

In the within session analysis, it was observed that with SSAE, combined EMG outperformed ($P < 0.001$) the sEMG using data of both healthy and amputee subjects. Similar performance was also repeated with LDA. However, SSAE achieved comparatively less error rates than LDA using both

combined and sEMG data as tabulated in table 1. It has been shown previously that iEMG from deep pronator and supinator muscles improves the performance in wrist pronation, wrist supination and opening/closing of terminal device [7, 42]. However, these studies were performed only with able-bodied subjects. Data of three amputee subjects were used in this study and it was found that combined EMG improved the performance for amputee subjects too. The reason is that EMG recorded from deep muscles through fine-wire and needle electrodes are less contaminated by crosstalk and hence could improve performance as compared to sEMG alone which suffers from many artifacts like electrode-skin interface and crosstalk from surrounding muscles etc.

In the between session analysis, long term performance of classifiers was accessed. Although combined EMG performed better than sEMG using both the classifiers (as shown in fig. 2). However, SSAE performed much better than LDA and achieved close error rates to that of within day analysis with a percentage point difference of 4.65 using combined and sEMG data of both healthy and amputee subjects. While this difference for LDA was 10.05. Hence, SSAEs maintained the performance to the data of seven sessions recorded with a gap of twenty-four hours and improved the performance as compared to LDA.

V. CONCLUSION

This study investigated whether recordings from deep pronator and supinator muscles could improve the performance of myoelectric control scheme for amputee subjects. The results showed that combined EMG significantly improved the performance than sEMG and further, SSAE proved to be more reliable than LDA for both the within and between sessions assessment of classifiers.

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